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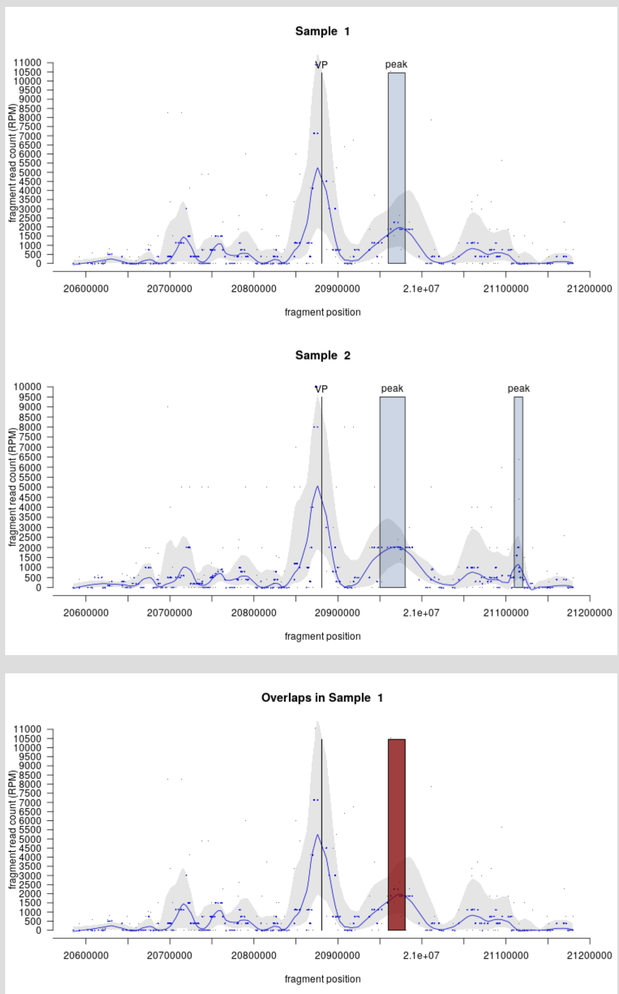
Basic4CVis: an R/Shiny app for 4C-seq quality control and interaction visualization

Introduction

Circular chromosome conformation capture with high-throughput sequencing (4C-seq) is a next-generation sequencing technique that allows to analyze three-dimensional interactions between one genomic region of interest (the experiment's viewpoint), and the remaining part of the genome [1]. The semi-quantitative, fragmented data structure of 4C-seq data requires adapted analysis strategies; simultaneous analysis of replicate data or samples from different conditions introduces additional challenges [2].

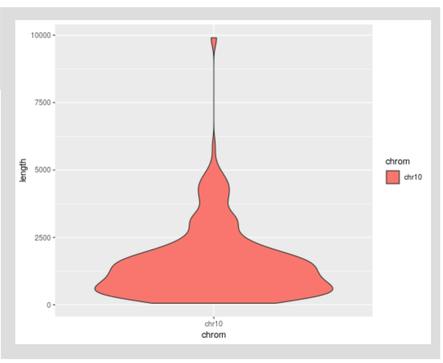
Methods & Results

Basic4CVis is an R/Shiny app that extends the R package Basic4Cseq [3], offering quality control routines, interactive visualization functions for different interaction types, and analysis options for sample groups.

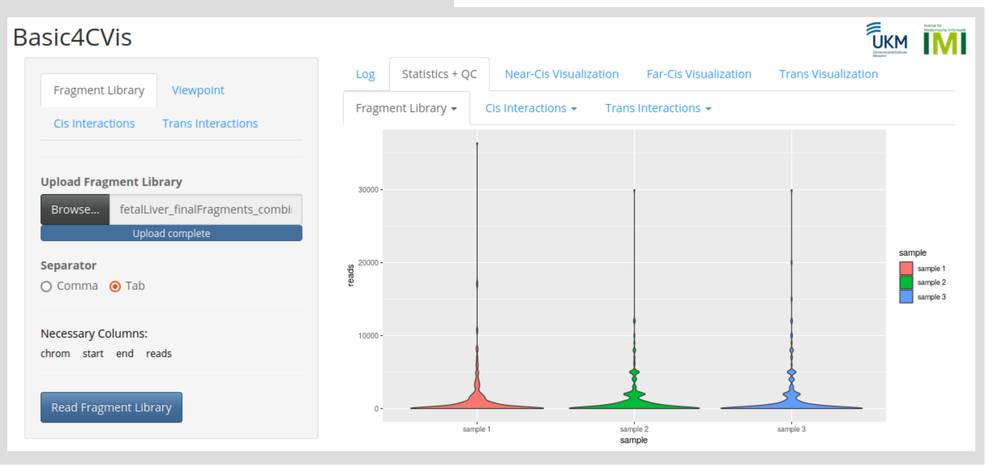


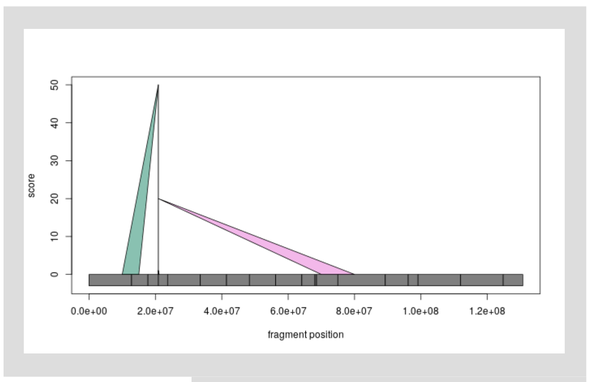
Near-cis Profiles

Statistics / QC



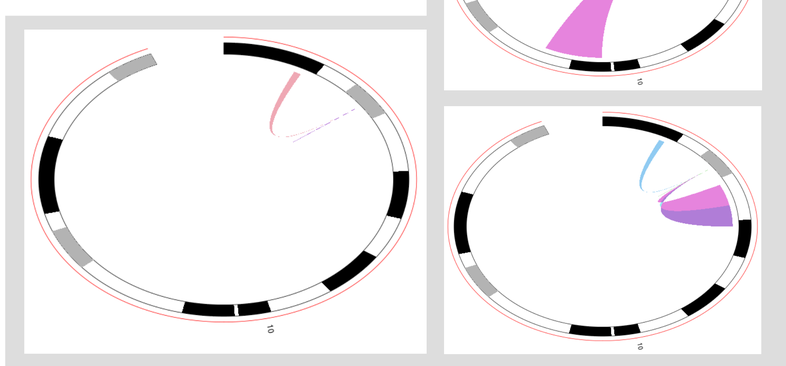
Main Shiny Application



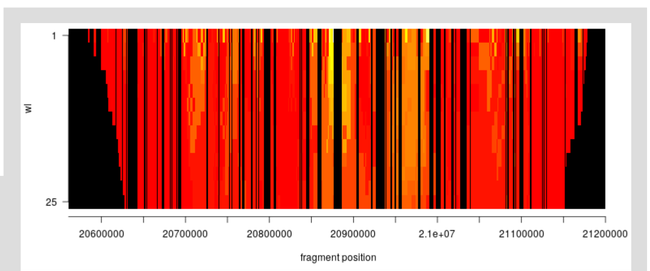


Far-cis Triangle Plot

Far-cis & Trans Circos Plots



Near-cis Heatmaps



Discussion & Conclusion

With its functionality for near-cis, far-cis and trans interaction visualization for single samples and sample groups originating from the same virtual fragment library, flexible import options and quality control functions, Basic4CVis is a helpful addition for the analysis of 4C-seq samples or sample groups.

References

[1] van de Werken et al. (2012) Robust 4C-seq data analysis to screen for regulatory DNA interactions, Nature Methods, 9, 969-972. [2] van de Werken et al. (2012) 4C technology: protocols and data analysis, Methods Enzymology, 513, 89-112. [3] Walter et al. (2014) Basic4Cseq: an R/Bioconductor package for analyzing 4C-seq data, Bioinformatics, 30, 3268–3269.

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